

Bill of Materials - VBF-T5R1NOCEILING-BOM-01

PDF release of the xlsx deliverable; values mechanically taken from tool outputs (cell/*_json, parameter set).

Sheet: BOM

Component	Mass (g/cell)	kg/kWh	Source / note
Positive electrode active material (NMC811)	31.846	0.8984	active volume fraction 0.665, density 3262 kg/m3 (parameter set); mass = layer_kg_m2 x area (cell/r4_A_calc_dfn.json)
Negative electrode active material (graphite)	20.580	0.5806	active volume fraction 0.75, density 1657 kg/m3 (parameter set); mass = layer_kg_m2 x area
Positive conductive additive (carbon black)	0.663	0.0187	ESTIMATE: literature default 2 wt% of positive coating solids (not in parameter set; excluded from simulation mass)
Positive binder (PVDF)	0.663	0.0187	ESTIMATE: literature default 2 wt% of positive coating solids (not in parameter set; excluded from simulation mass)
Negative conductive additive (carbon black)	0.429	0.0121	ESTIMATE: literature default 2 wt% of negative coating solids (not in parameter set; excluded from simulation mass)
Negative binder (CMC/SBR)	0.429	0.0121	ESTIMATE: literature default 2 wt% of negative coating solids (not in parameter set; excluded from simulation mass)
Separator	0.259	0.0073	porosity 0.47, density 397 kg/m3 (parameter set); mass = layer_kg_m2 x area
Electrolyte (1 M LiPF6 in EC:EMC 3:7 w/w)	11.565	0.3263	ESTIMATE: pore volume 9.64 mL x literature density 1.2 g/cm3, fill factor 1.0 (not in parameter set; excluded from simulation mass)
Positive current collector (Al, 6 um)	2.496	0.0704	density 2700 kg/m3 (parameter set); mass = layer_kg_m2 x area
Negative current collector (Cu, 4 um)	5.521	0.1558	density 8960 kg/m3 (parameter set); mass = layer_kg_m2 x area
Enclosure (can/pouch)			Not modeled (beyond pure simulation boundary)
Tabs			Not modeled (beyond pure simulation boundary)
SUMMARY: total mass, simulation contract caliber (electrolyte excluded)	60.703	1.7125	cell/r4_A_calc_dfn.json:mass_kg; formula = sum layer thickness x (1-porosity) x density x area
SUMMARY: total mass, industry caliber incl. electrolyte + additive/binder estimates	74.452	2.1004	estimate (electrolyte + additive/binder literature defaults added to contract mass)
SUMMARY: cell energy	35.446 Wh = 0.035446 kWh		cell/r4_A_calc_dfn.json:energy_wh (time integration of V x I at 5 A)